Listen Before You Classify: A DSP Front End and an Adapted Classifier for Efficient, Reliable Sound Alerts on Wearables

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***Abstract*—Wearable sound-awareness devices for Deaf and hard-of-hearing people run a neural audio classifier on every half second of microphone input, which costs energy, and a general-purpose classifier misses or confuses the few sounds that matter. We combine a classical signal-processing front end with an adapted classifier for five alert classes (fire/smoke alarm, siren, car horn, baby crying, door). A Butterworth high-pass filter, short-time band energies and a fast and a slow minimum-tracking noise floor wake the classifier only when a new sound starts; a Goertzel resonator bank with a temporal-pattern matcher detects standard alarm tones; a veto removes the classifier's typical siren/fire-alarm confusion. The classifier is a small head trained on frozen embeddings of YAMNet or the mobile EfficientAT-mn10, with per-class score smoothing and thresholds set under a false-alarm budget on tuning data only. On 8 h of test streams (480 real and standard alarm events in six unseen environments) the system matches the recall of the deployed always-on YAMNet (44 % vs. 42 %) with ten times fewer false alarms (1.8 vs. 16.8 per hour), detects 13 points more than YAMNet at the same false-alarm rate ($p < 10^{-7}$), and runs the classifier 7 % of the time in quiet audio. Looser budgets reach 62 % recall on the benchmark, but on naturally recorded home audio they multiply false alerts by ten: only the strictest setting transfers, where the system raises fewer false alerts in a real home (0.36 vs. 0.55 per hour) and almost no fire alarms on look-alike beeps (0–2 % vs. 4–12 %). We argue that benchmark-tuned sound alert systems must be validated on real audio before deployment.**

***Index Terms*—sound event detection, wake-up detector, Goertzel algorithm, noise-floor tracking, transfer learning, assistive technology**

I. Introduction

Wearable sound-awareness devices turn important environmental sounds into vibration and visual alerts for Deaf and hard-of-hearing (DHH) people [1], who rank safety alarms, sirens, vehicle horns, a crying baby and someone at the door among the sounds they most want to know about [2]. They typically run a pretrained classifier such as YAMNet [3] on every half second of audio. This has two costs. *Energy*: the classifier runs although nothing new happens most of the time, and battery drain was a key constraint for on-watch recognition [1]. *Reliability*: a general-purpose classifier confuses related safety classes, misses standardised signals such as the ISO 8201 temporal-three (T-3) smoke-alarm pattern [4], and raises false alerts that erode trust.

We ask: *can a wearable catch the sounds that matter more reliably, with fewer false alarms, while running its classifier only a small fraction of the time?* Our answer combines classical DSP where the problem is well-specified (when does a new sound start, is this a standard alarm tone) with light adaptation of a pretrained classifier where it is not. Contributions:

1) A training-free DSP front end (Butterworth high-pass, short-time band energies, a fast and a slow minimum-tracking noise floor, a Goertzel resonator bank with a T-3 matcher, a siren veto) costing 0.5 % of one classifier inference, which keeps the classifier asleep 87–93 % of the time in quiet audio and adapts to rooms where a fixed level threshold does not.
2) An adapted classifier for five alert classes: a linear head on frozen embeddings (YAMNet or EfficientAT-mn10 [5]), trained on public data with strict separation of uploaders and recordings from the test set, with per-class score choice, temporal smoothing and thresholds set under a false-alarm budget.
3) A recall–false-alarm evaluation on 480 test events in six unseen environments: at every false-alarm rate the system detects 12–14 points more than YAMNet at the same false-alarm rate; at the strictest budget it matches the deployed app's recall with ten times fewer false alarms.
4) Evidence of a transfer gap: operating points tuned on mixed benchmark streams raise up to ten times more false alerts on naturally recorded home audio (SINS, DESED, real look-alike beeps); only the strictest one transfers.
5) Diagnostics that explain absolute numbers: synthetic standard alarms overstate DSP detectors, and a large share of the test clips is not recognised even without background noise. All data are public; the study runs in Colab.

II. Related Work

Sound awareness for DHH users. SoundWatch [1] classified environmental sounds on a smartwatch and found battery drain a key concern; ProtoSound [6] personalised recognition from few examples. Neither reduces *when* the classifier runs, nor reports false alerts in real homes.

Wake-up detectors and cascades. Always-on systems place a low-power detector before a costly stage [7]; keyword spotting wakes large models with small ones [8], and a recent cascade screens audio before a trimmed YAMNet [9]. Our

gate is training-free and follows the room through noise-floor tracking, a simplified form of minimum statistics [10]; we test an ordered-statistic CFAR threshold [11] as an ablation.

Efficient audio tagging and adaptation. EfficientAT distils transformer taggers into MobileNetV3 CNNs and outperforms YAMNet on AudioSet [12] at similar cost [5]. Training a light head on frozen embeddings is the standard transfer recipe; we add per-class selection and smoothing under an explicit false-alarm budget, and test whether the budget holds on real audio.

III. Method

A. Signals, classes and alert rule

Audio is sampled at $f_s = 16\,\text{kHz}$. The Nyquist frequency of 8 kHz covers the piezo smoke-alarm band (about 2.8–3.6 kHz) and the 520 Hz low-frequency alarm [13] with its odd harmonics up to the 13th. A classifier window is 0.975 s, every 0.5 s. Five alert classes, in priority order: fire/smoke alarm, siren, car horn, baby crying, door (knock or doorbell, one alert). A class fires when its score reaches its threshold in N classified windows in a row ($N = 2$ for alarm, siren and baby, 1 for horn and door; a run survives skipped windows if classified windows are at most 2 s apart), with a 20 s cooldown; the most urgent class wins.

B. Gate path

High-pass filter. Butterworth, order 2, $f_c = 100\,\text{Hz}$ ($-3\,\text{dB}$), one biquad

$$H(z) = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2}}{1 + a_1 z^{-1} + a_2 z^{-2}}, \tag{1}$$

with a double zero at $z = 1$ and two poles inside the unit circle (Fig. 2). It removes DC, rumble and handling noise below the lowest target component.

Band energies and noise floors. Each 20 ms frame ($N = 320$, Hann) is transformed with a 512-point FFT and its power summed in $B = 16$ log-spaced bands (100 Hz–7.6 kHz), smoothed with a one-pole low-pass ($\tau = 60\,\text{ms}$). With $E_b[t]$ the band level in dB, each band's floor follows the minimum with a bounded rise,

$$F_b[t] = \min\big(E_b[t],\, F_b[t-1] + \rho T\big), \qquad \rho = 3 \text{ dB/s}: \tag{2}$$

it drops to a quieter background at once and absorbs a steady sound within seconds. A sound that swells more slowly than ρ (a siren fading in) drags this floor up with it, so a second floor $S_b[t]$ with $\rho_s = 0.3\,\text{dB/s}$ remembers how quiet the room was. The novelty

$$\nu[t] = \max\Big(\max_b\big(E_b - F_b\big),\ \max_b\big(E_b - S_b\big) - \delta\Big)[t], \tag{3}$$

with $\delta = 6\,\text{dB}$, is the largest rise of any band above its own floors. If the maximum of ν over the 0.5 s just received exceeds Δ, the classifier runs on the window that ends now, and stays awake for a hangover H. Since the window spans the last 0.975 s, the onset that woke the gate lies inside it: waking adds no delay.

C. Alarm path and siren veto

Goertzel resonator bank. The power of DFT bin k in a frame is computed by the Goertzel recursion [14]

$$s[n] = x[n] + 2\cos(\omega_k)\, s[n-1] - s[n-2], \tag{4}$$

$$|X_k|^2 = s_1^2 + s_2^2 - 2\cos(\omega_k)\, s_1 s_2, \tag{5}$$

a second-order IIR resonator with poles at $e^{\pm j\omega_k}$ on the unit circle, read once per frame: 17 probes from 2.8 to 3.6 kHz (50 Hz apart) and 520, 1560, 2600 Hz. A Goertzel probe costs one multiply per sample, so it beats a full FFT only below about 14 probes; an implementation could read these bins from the gate's FFT instead.

Tone track and pattern. A frame is "on" when the strongest probe holds more than 15 % of the frame energy (35 % for the square-wave probes together) and is within 8 dB of its 1.5 s running peak, which rejects reverberant tails. After closing 100 ms dropouts, an alarm is declared on three pulses of 0.3–0.8 s separated by gaps of 0.3–0.8 s (the T-3 timing with tolerance). Band, share and rule were selected on the design half of real recordings (Sec. IV-C).

Siren veto. YAMNet also scores *Alarm* on sirens, and the alarm class has priority; a classifier alarm is therefore accepted only when its score exceeds the siren score in the same window.

Knock path. An impulse-rhythm detector (spectral flux peaks with regular inter-onset intervals) was searched on tuning data like every other setting. It raises stock YAMNet's door recall at strict budgets (3 % to 19 % at $b = 0.1$) but adds no detection to the trained classifier at equal false alarms, so the final system does not use it.

D. Adapted classifier

Backbones. YAMNet (MobileNetV1, 521 classes, $M_Y = 70\,744\,064$ multiply-accumulates (MACs) per window, counted from its convolution and dense layers) and EfficientAT-mn10 (MobileNetV3, 527 classes, 32 kHz input, 62 897 600 MACs per window), both frozen. A backbone's own score for a class is the maximum over that class's labels.

Head. Multinomial logistic regression or a one-hidden-layer MLP on the standardised embedding (1024-d for YAMNet, 960-d for mn10), optionally with the logits of the backbone's own class scores appended, over the five classes plus "other". A training window is labelled with a class when that target is sounding (frames within 30 dB of its own peak) for at least 60 ms of it. Model and regularisation are chosen by mean average precision on the tuning windows. The head costs about 7 000 MACs.

Per-class choice, smoothing and thresholds. For each class, candidate scores are the backbone's own score, the head's probability and their mean, each smoothed by a causal moving average over $k \in \{1, 2, 3\}$ windows. Each candidate's threshold is the lowest whose false alarms on the tuning streams stay within a budget b per hour; among candidates within budget, the one with the highest tuning recall is kept. We restrict candidates to one backbone so that a single network runs.

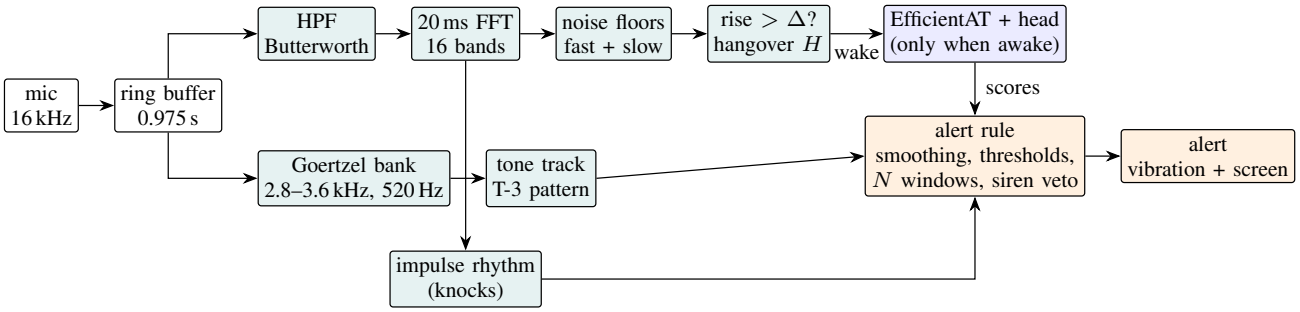


Fig. 1. The system. The gate path wakes the classifier, which reads the last 0.975 s from the ring buffer; the alarm path raises alerts by itself; all alerts pass one alert rule.

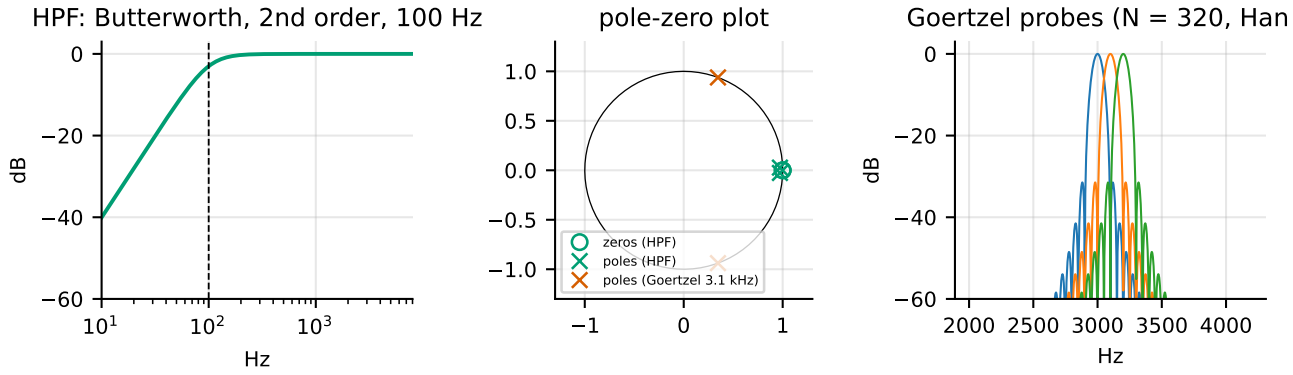


Fig. 2. DSP design: high-pass filter magnitude response, pole-zero plot (HPF and one Goertzel resonator) and Goertzel probe responses ($N = 320$, Hann).

E. Cost

Per 0.5 s hop the gate costs about 195 000 MACs and the alarm path 177 500, together 0.53 % of one YAMNet inference. Compute relative to always-on YAMNet is $(d\,M + M_{\text{gate}} + M_{\text{alarm}})/M_Y$, with d the classifier duty (fraction of windows classified) and M the chosen backbone's cost. We count MACs, not battery energy (Sec. VI).

IV. Experimental Setup

A. Streams and data

Following DCASE 2017 Task 2 [15], we mix events into 5-minute DEMAND [16] recordings (background -50 to -35 dBFS). Each event stream holds 10 targets (2 sirens, 2 car horns, 2 baby cries, 2 door sounds, one ISO 8201 synthetic alarm (piezo or 520 Hz) and one real smoke/fire-alarm recording) at an SNR from $\{-5, 0, 5, 10, 20\}$ dB over the event's active frames, and 8 distractors (0–20 dB) that never overlap a target, half of them easily confused with targets (bells, phones, impacts, beeps, speech, music, traffic, microwave and reversing beepers). Each has an *idle* twin with the background only. Targets and distractors come from FSD50K [17] and ESC-50 [18] (official splits) and from Freesound [19] (baby cries, real alarms and look-alike beeps, split by uploader). *Test*: 6 environments (living room, office, cafeteria, street, bus, park; 4.0 h of event and 4.0 h of idle streams, 480 targets, 96 per class). *Tuning*: 11 other environments (7.3 h + 7.3 h), 30 % of the tuning clips. *Training* (heads only): the tuning environments' other microphones, the other 70 % of the tuning clips, plus 597 Freesound recordings titled as target sounds and 1 291 UrbanSound8K [20] horn and siren excerpts, excluding every uploader and recording of the FSD50K evaluation set, the ESC-50 test fold and the held-out Freesound half (29.3 h of training streams, 210 848 windows).

B. Calibration

Gates are calibrated as in our earlier protocol: the lowest-duty setting (Δ or level T, and $H \in \{0, 1, 2\}$ s) whose *gate-caused* miss rate (detections of the same always-on classifier lost because the gate slept) stays within 5 %. Seven gate statistics were compared on tuning data (Table III). Class thresholds are set under false-alarm budgets $b \in \{0.1, 0.3, 1, 3\}$ per hour per class, for stock YAMNet as well, so classifiers are compared at equal tuning false alarms.

C. Real recordings

Alarms. 161 Freesound recordings: 38 smoke alarms, 30 fire alarms (bells, horns, beeps) and 93 look-alikes (microwave and oven timers, reversing beepers, alarm clocks, monitors, phones), split by uploader into a design half (used to select the alarm detector) and a held-out half. Each held-out clip is tested clean and mixed into a test-environment background at 10 and 0 dB after 5 s of that background. **Stress.** Synthetic alarms outside ISO 8201 (pitch, timing, reverberation) and look-alikes

(a T-4 carbon-monoxide pattern, a continuous 1 Hz beeper, a single triple beep), 20 each at 5 dB. **Homes.** DESED public evaluation [21]: 692 real 10 s home recordings with 2 765 labelled events of 10 classes. SINS [22], [23]: one real home, 1 981 10 s segments (5.5 h) labelled with the household's activity, none containing our target classes, so every alert is false. Nothing is tuned on these data except, explicitly, the choice of recommended budget (Sec. V-C).

D. Metrics

A target is detected if an alert of its class fires between its onset and 1 s after its end. We report recall with 95 % bootstrap intervals over streams, false alarms per hour (event streams, idle streams), classifier duty and compute, and exact McNemar tests on paired events. As FSD50K labels are clip-level, we also report recall on the 348 test events whose FSD50K raters marked the target as *present and predominant* (all non-FSD50K events kept), a subset fixed by the dataset's annotators.

V. Results

A. Recall and false alarms on the benchmark

Table I and Fig. 3 give the test results. Tuned to a false-alarm budget, stock YAMNet trades most of its recall for fewer false alarms (31 % at 1.8 per hour). The DSP alarm path recovers the standard alarms YAMNet misses (alarm recall 7 % to 50 % at $b = 0.1$) and the gate lets YAMNet sleep 93 % of the time in quiet audio. Adapting the classifier adds most for the classes its stock labels describe poorly: with the mn10 head, door recall rises from 18 % to 43 % and horn from 16 % to 25 %. At the same tuning false-alarm rate as tuned YAMNet, the system detects 13 points more (98 vs. 35 events detected by only one of the two, $p = 4 \cdot 10^{-8}$); against the deployed app, recall is the same (44 % vs. 42 %, $p = 0.51$) with ten times fewer false alarms. At matched false-alarm rates the adapted system lies 12–14 points above YAMNet along the whole curve, and looser budgets reach 62 % (72 % on the predominant subset) at 10 false alarms per hour. The strict budget costs sirens most (21 %): their head scores rarely clear the very high thresholds the budget demands.

Which classifier. Table II compares backbones by window-level average precision on the test streams. A head on YAMNet's embedding helps alarms and doors but not sirens and horns, which YAMNet's own labels, trained on far more data, already separate. The mn10 embedding gives the best head on every class but horn. mn10's own label scores saturated in our pipeline (near 1 for every class), so it is used only through its head; mn04 (12.7 M MACs) was weaker and is not used.

B. The gate

Duty cycling cannot save anything: classifying every second window already loses more than 5 % of YAMNet's detections on tuning data. Calibrated to YAMNet at the 5 % budget, the two-floor gate lets it sleep 87 % of the time in idle test streams at 41.5 % vs. 42.3 % recall; the fixed level gate sleeps 79 %. A fixed threshold depends on how loud a room is: its idle

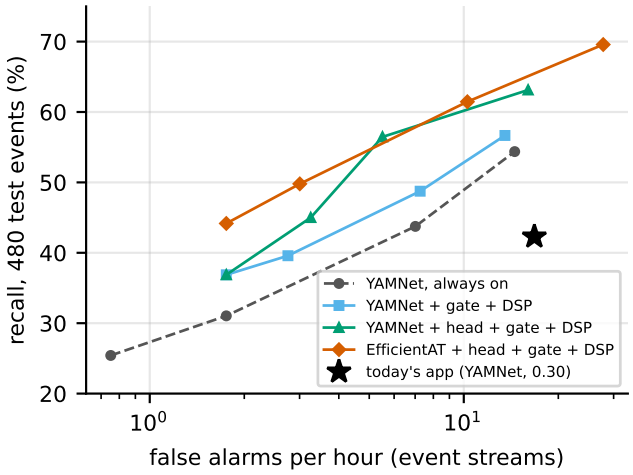


Fig. 3. Recall against false alarms per hour on the test event streams, for false-alarm budgets of 0.1–3 per hour per class set on tuning data. Star: the deployed app.

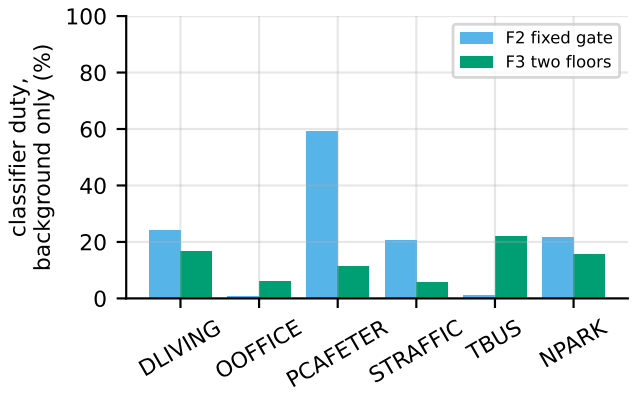


Fig. 4. Classifier duty in background-only audio per unseen test environment (YAMNet, 5 % budget).

duty ranges from 0.8 % (office) to 59 % (cafeteria), while the two-floor gate stays between 5.8 % and 22 % in all six unseen rooms (Fig. 4). Longer smoothing, a two-band rule and an ordered-statistic CFAR threshold did not improve on the one-floor statistic; CFAR was clearly worse because its reference median rises during a long sound just when the classifier's scores peak. The slow second floor helped at every budget (Table III). For the final system the gate is recalibrated against its own detections ($\Delta = 24$ dB, $H = 1$ s) and the classifier runs 6.8 % of the time in idle streams; the gate and alarm path together add 13 detections and lose 5 against the same classifier always on.

C. Transfer to naturally recorded audio

The benchmark budget does not transfer (Table IV, Fig. 5). At $b = 1$, the operating point with the best benchmark recall, the system finds many more real smoke alarms than the app (57 % vs. 22 % at 0 dB) but raises a fire alarm on a third of

TABLE I

TEST RESULTS (6 UNSEEN ENVIRONMENTS, 480 TARGETS, 96 PER CLASS). THRESHOLDS SET ON TUNING DATA UNDER A FALSE-ALARM BUDGET b (PER HOUR PER CLASS); "APP" IS THE DEPLOYED RULE (YAMNET, 0.30). FA/H ON EVENT / IDLE STREAMS; COMPUTE RELATIVE TO ALWAYS-ON YAMNET IN IDLE STREAMS; "PRED." = RECALL ON THE PREDOMINANT-TARGET SUBSET.

System	b	Recall (%) [95 % CI]	Alarm	Siren	Horn	Baby	Door	FA/h ev.	FA/h idle	Compute (%)	Pred. (%)
YAMNet always on (app)	–	42.3 [37.5, 47.1]	31	58	33	67	22	16.8	0.25	100	49.4
YAMNet always on, tuned	0.3	31.0 [27.1, 35.0]	10	30	16	81	18	1.8	0	100	38.8
YAMNet + gate + alarm DSP + veto	0.1	36.9 [33.3, 40.6]	50	30	8	77	19	1.8	0	7.3	47.1
YAMNet head + gate + DSP	0.1	36.9 [33.5, 40.2]	52	27	14	77	15	1.8	0	5.4	47.7
mn10 head + gate + DSP	**0.1**	**44.2 [40.2, 48.3]**	56	21	25	76	43	**1.8**	**0**	**6.5**	**53.2**
mn10 head + gate + DSP	0.3	49.8 [46.0, 54.0]	60	33	31	78	46	3.0	0.25	11.1	58.0
mn10 head + gate + DSP	1	61.5 [57.7, 65.2]	74	53	40	84	56	10.3	0.50	19.7	72.1

TABLE II

WINDOW-LEVEL AVERAGE PRECISION ON THE TEST STREAMS: BACKBONE'S OWN LABELS / TRAINED HEAD.

Backbone (MACs)	Scores	Mean	Alarm	Siren	Horn	Baby	Door
YAMNet (70.7 M)	labels	.41	.38	.53	**.20**	.74	.22
	head	.50	.71	.49	.14	**.79**	.36
mn04 (12.7 M)	labels	.46	.59	.54	.18	.69	.28
	head	.47	.65	.48	.15	.64	.43
mn10 (62.9 M)	head	**.57**	**.75**	**.56**	.23	**.79**	**.51**

TABLE III

CLASSIFIER DUTY (%) ON THE TUNING STREAMS WITHIN EACH GATE-CAUSED MISS BUDGET (YAMNET).

Gate	1 %	3 %	5 %
Every k-th window	100	100	100
Fixed level	60.7	51.6	38.6
One floor, max band rise	44.3	33.2	28.4
+ 150 ms smoothing, two bands	44.4	31.9	26.4
+ OS-CFAR	–	61.2	42.9
Two floors, $\delta = 6$ dB (selected)	**43.8**	**29.9**	**24.4**
$\delta = 10$ dB	45.0	31.3	27.1
$\delta = 14$ dB	44.5	33.5	26.7

real look-alike beeps and on almost every synthetic look-alike, and ten times more false alerts in a real home (5.3 vs. 0.55 per hour; 11.7 with the TV on, 16.9 during social activity). The streams' backgrounds and distractors do not contain enough of what a home contains: television, conversation, appliance beeps. Only $b = 0.1$ keeps real-home false alerts at or below the app's, and it nearly eliminates look-alike alarms; its smoke-alarm detection equals the app's, and it detects 520 Hz alarms the app never does (95 % vs. 0 %). Its cost is fire bells and horns, which fall from 31–62 % to 8–23 %. We recommend $b = 0.1$; because this choice uses SINS, the SINS figure for it is not an independent test.

Compute in real homes. With the recommended system the classifier runs under 1 % of the time in an empty home, 9 % while working, 31 % while cooking and 73 % with the TV on; over the 5.5 h it runs 36 % of the time, about two thirds less compute than always-on YAMNet. On DESED's labelled events the gate was awake for 94 %, lowest for steady noises such as a vacuum cleaner, which the floors absorb.

TABLE IV

REAL RECORDINGS: TODAY'S APP VS. THE MN10 SYSTEM AT THREE BUDGETS. ALARM ROWS: SHARE RAISING A FIRE ALARM (%), CLEAN / 10 DB / 0 DB. ↓: LOWER IS BETTER.

	App	$b = 0.1$	$b = 0.3$	$b = 1$
Smoke alarms (23)	43/43/22	39/35/26	43/39/30	70/61/57
Bells, horns (13)	62/62/31	15/23/8	46/38/15	54/54/46
Look-alikes (50) ↓	12/6/4	**2/2/0**	12/10/6	36/32/28
T-3 3.1 kHz/520 Hz	75/0	85/95	90/100	100/100
Synth. look-alikes ↓	48	50	82	98
SINS alerts/h ↓	0.55	**0.36**	1.64	5.27
SINS duty (%)	100	36	41	40
DESED wake (%)	–	94	95	98

D. Alarms: synthetic versus real

On synthetic ISO 8201 alarms a tone-and-pattern detector looks excellent; real recordings tell a different story. Characterising the 32 design-half real alarms with DSP (FFT peak of tonal frames, beep timing) shows a median pitch of 3.0 kHz but only 54 % within the 2.8–3.6 kHz probes (range 1.0–3.5 kHz), and only 6 of 32 follow the T-3 pattern: 4 are continuous tones, 3 other rhythms and 19 are not narrowband tones at all (bells, horns, distorted recordings). The DSP path is a safety net for standard tones in noise, not a replacement for the classifier.

VI. DISCUSSION

What each part buys. The gate buys compute without losing detections and adapts to the room. The alarm path and the veto fix the classifier's two worst errors on alarms. The adapted classifier buys recall at a given false-alarm rate, mostly for doors and alarms. None of these needs on-device training.

Benchmarks are not homes. Mixed streams are indispensable for controlled comparisons, but a false-alarm budget set on them reflects their distractors, not a living room. Adapting the classifier made this gap larger, not smaller: a head trained to say "alarm" to beeps says it to the microwave too. Sound-alert systems should report false alerts on naturally recorded home audio, per activity, next to benchmark recall.

Why recall is moderate. 38 % of test events are at or below the background level, and the clips are hard: with no background at all, a YAMNet-based classifier of our pipeline recognises only 63 % of the test clips once (74 % of the predominant subset). Real alarms keep sounding and visitors

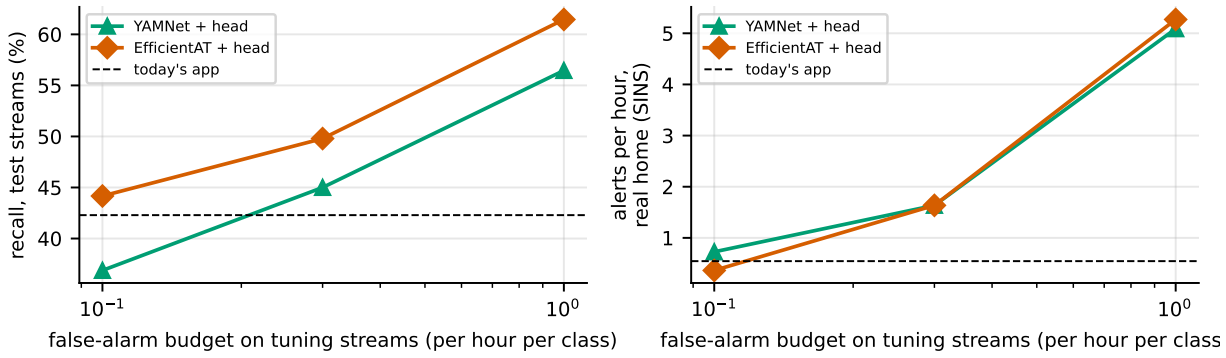


Fig. 5. The transfer gap. Left: recall on the test streams at each false-alarm budget. Right: alerts per hour in 5.5 h of a real home (SINS), where every alert is false. Dashed: the deployed app.

knock again, so recall per event understates detection over an episode.

Limitations. Real smoke-alarm recordings are scarce (38), so their differences are not significant. The recommended budget was chosen with SINS, one home. Sirens and fire bells lose recall at the strict budget. We count MACs, not battery energy: the microphone, conversion and the front end's own FFT run continuously, so the battery saving is smaller than the classifier saving. mn10's own labels could not be used; carbon-monoxide alarms were left out for lack of public recordings.

VII. Conclusion

A classical front end costing 0.5 % of one neural inference lets a wearable's classifier sleep most of the time and detects standard alarm tones the classifier misses; a light head on a mobile audio embedding adds recall where stock labels are weak. Together they match the deployed classifier's recall with ten times fewer false alarms on a benchmark, and raise fewer false alerts in a real home. Pushing for more benchmark recall multiplied real false alerts: alert systems for DHH users should be tuned and validated on real home audio. Future work: continuous labelled home recordings with real safety events, hard negatives from homes, and on-device energy measurement.

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